CrossPiezo: Databases Agree on Promising Regions but Cannot Resolve the Elite Tail in Piezoelectric Screening

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Abstract

High-throughput piezoelectric databases are widely used to train and benchmark machine-learning models, yet their cross-source agreement is usually reported as a single global correlation. Here we reframe cross-database comparison as a question of *screening resolution*: do the Materials Project (MP) and JARVIS databases agree on broad promising regions of chemical space, yet fail to reproducibly resolve the elite tail of candidates? Using the frozen CrossPiezo-Invariant-v1 benchmark of 573 strict structure-matched pairs (P0) and 207 tighter matches (P2), we compute coordinate-invariant response scalars F1, F3 and F4 and sweep the screened quantile from 1% to 50%. The normalised area under the chance-adjusted concordance curve (nAUCC) for F1 on P0 is 0.105, and the simultaneous 95% lower confidence bound of adjusted overlap first exceeds 0 at $q = 33\%$; no five-quantile persistent onset is observed at $\delta = 0.05$, showing that even chance-adjusted agreement is not reliably sustained in the elite tail. The high-response subset and the tighter P2 panel show even weaker tail resolution, confirming that the disagreement is not a simple sample-size artifact. Simpler scalar controls (unit-cell volume, band gap) are markedly more cross-source consistent than the piezoelectric response, with positive $\Delta\tau$ and ΔnAUCC. Source-robust portfolios improve worst-source recall and NDCG relative to JARVIS-only or MP-only selection on a frozen holdout fold, but this is two-source decision robustness, not independent physical validation. All numbers are hash-bound and traceable to the Phase 7B manifest.

1 Introduction

Piezoelectric response tensors are central to electromechanical materials design, yet costly to compute with density-functional perturbation theory (DFPT). The Materials Project (MP) and JARVIS projects have released thousands of computed tensors [1, 2], and recent equivariant neural networks report strong in-source prediction accuracy [3, 4, 5]. Improved agreement with a held-out subset of one database does not guarantee that a candidate is robustly ranked across independently generated high-throughput datasets [6, 7].

We introduce *screening resolution* as a diagnostic lens. Rather than asking "what is the average rank correlation between two sources?" we ask "at what elite fraction q does the overlap between source-specific top-q sets become statistically distinguishable from chance?" This distinction matters for materials discovery: a model that is validated only on aggregate correlation may still disagree with an independent source on the very candidates a human experimenter would pursue.

2 Screening-resolution framing

Paired universe. We use the frozen Phase 6A panel: P0 ($n = 573$) and P2 ($n = 207$), structure-matched pairs with identical or near-identical crystal structures across MP and JARVIS. We do not modify panel membership.

Response functionals. For each source we compute three coordinate-invariant scalars: F1 = Cartesian Frobenius norm $\|\mathbf{e}\|_F$; F3 = true collinear longitudinal maximum $\max_{\|\boldsymbol{n}\|=1} |n_i e_{ijk} n_j n_k|$ [8]; F4 = Kelvin/Mandel operator norm on symmetric strain.

Chance-adjusted concordance curve. For each panel, metric and quantile $q = 1\%, \ldots, 50\%$, we report the observed top-q Jaccard overlap J_q, the exact hypergeometric null expectation $\mathbb{E}[J_q]$ under independent rankings, and the chance-adjusted overlap

$$\widetilde{J}_q = \frac{J_q - \mathbb{E}[J_q]}{1 - \mathbb{E}[J_q]}. \tag{1}$$

We construct a simultaneous 95% confidence band around $\widetilde{J}_q$ using a paired bootstrap with a studentized sup-norm critical value [9, 10]. The normalised area under the concordance curve is

$$\text{nAUCC} = \frac{1}{q_{\max} - q_{\min}} \int_{q_{\min}}^{q_{\max}} \widetilde{J}_q \, dq, \tag{2}$$

and the persistent consensus onset is the smallest q for which the lower band exceeds a threshold δ for at least five consecutive quantiles.

Controls and heterogeneity. We compare piezoelectric response against volume, band gap, energy above hull and dielectric trace, recording source fields, common N and available N. We report $\Delta\tau = \tau_{\text{control}} - \tau_{\text{piezo}}$ and ΔnAUCC with grouped paired-bootstrap confidence intervals, permutation p-values and Benjamini–Hochberg FDR. Heterogeneity is analysed by crystal system, anion category, response-magnitude quartile, match-quality terciles and heavy-atom fraction, with subgroup minima $N = 30$.

Robust portfolio benchmark. We fix an elite fraction $q^* = 10\%$ and budgets $b \in \{1.0, 1.5, 2.0\} \times \lfloor q^* n \rfloor$. Strategies include single-source, average percentile, Borda count, maximin percentile, intersection-first, balanced union, disagreement abstention and a greedy minimax oracle. We evaluate worst-source recall, worst-source NDCG (with the ideal computed from the full universe), minimax regret and coverage on a deterministic holdout fold obtained by hashing `pair_id` modulo 5.

3 Results

3.1 Screening-resolution curves

Table 1 summarises the concordance diagnostics. On P0 the three functionals behave similarly: nAUCC is 0.105 for F1, 0.110 for F3 and 0.106 for F4. The pointwise lower confidence bound first exceeds 0 at 33% for F1, but a five-quantile persistent onset at $\delta = 0.05$ is not observed for any primary metric, and at $\delta = 0.10$ it is not observed for F1. Thus, even if one screens half of the universe, the two sources only barely agree more than would be expected by chance.

Table 1: Screening-resolution summary for P0 ($n = 573$) and P2 ($n = 207$). $q_{0.00}$ is the pointwise first crossing; $q_{0.05}^{\text{persist}}$ requires five consecutive lower-confidence bounds above 0.05.

Panel	Metric	nAUCC	$q_{0.00}$	$q_{0.05}^{\text{persist}}$
P0	F1_Frobenius	0.105	33%	–
P0	F3_Longitudinal	0.110	20%	38%
P0	F4_KelvinOp	0.106	28%	42%
P2	F1_Frobenius	0.060	–	–
P2	F3_Longitudinal	0.043	–	–
P2	F4_KelvinOp	0.046	–	–

The high-response subsets (top 50% by pooled F1) and the tighter P2 panel show *negative* nAUCC values, meaning the elite tail is less reproducible than random ranking. This confirms that the elite-tail resolution limit is a robust feature of the data, not an artifact of averaging over low-response materials.

3.2 Scale, order and tail

Source-wise quantile normalization leaves Kendall τ essentially unchanged (by monotonic invariance), so the disagreement is not a simple monotonic calibration error. ECDF Kolmogorov–Smirnov tests and quantile-mapping residuals reveal genuine scale shifts, while top-10% threshold-crossing counts remain large, indicating that the tail itself is unstable even after an order-only calibration.

3.3 Property controls

Table 2 shows that all available controls are more cross-source consistent than piezoelectric F1. Volume is the strongest control (P0 $\tau = 0.967$, nAUCC $= 0.918$), followed by band gap ($\tau = 0.908$, nAUCC $= 0.897$). All controls have positive $\Delta\tau$ and ΔnAUCC with respect to F1. The elastic bulk modulus is available only from MP in this processed release, so it is marked `insufficient_N` for cross-source comparison.

Table 2: Property-control comparison on P0. Positive Δ means the control is more consistent than F1.

Attribute	τ	nAUCC	$\Delta\tau$	ΔnAUCC	common N
F1_Frobenius	0.245	0.105	–	–	573
volume	0.967	0.918	0.722	0.813	573
band_gap	0.908	0.897	0.663	0.792	573
energy_above_hull	0.611	0.533	0.366	0.428	573
dielectric_trace	0.482	0.286	0.236	0.181	573

3.4 Electronic/ionic decomposition

Convention-complete electronic/ionic/total tensor decompositions are available for fewer than the required 100 pairs in both sources, so the decomposition analysis is marked `insufficient_N`. We therefore do not draw conclusions about ionic versus electronic contributions.

3.5 Heterogeneity

Subgroups with $N \geq 30$ show variation around the panel-average τ, but after FDR correction most subgroup differences are not statistically significant. Continuous match-quality covariates (RMS distance, lattice distance) are weakly correlated with absolute rank displacement, confirming that the residual disagreement is not dominated by a single match-quality variable.

3.6 Robust portfolio benchmark

On the frozen holdout fold, source-robust strategies improve worst-source recall and NDCG relative to JARVIS-only or MP-only selection; the balanced-union strategy reaches a worst-source recall of 1.0 at budget factor 2.0 on P0 F1, while the single-source strategies remain below 0.30 at the same budget. Disagreement abstention tunes its penalty on the development fold and reduces tail regret by penalising high cross-source disagreement. Full benchmark numbers are given in the hash-bound `portfolio_benchmark.csv` artifact.

4 Discussion

The weak elite-tail resolution is a statement about database definitions and processed fields, not about the physical correctness of either source. Simpler, more convention-robust observables (volume, band gap) are reproduced far more reliably than the piezoelectric tensor, suggesting that a large fraction of the disagreement stems from workflow-specific conventions rather than from irreducible physics.

Source-robust portfolios improve the worst-case recall of a two-source decision maker, but they do not validate physical truth. A third computational protocol or an experimental adjudication set would be required to identify which source, if either, is "correct" on any specific candidate.

5 Data availability

Hash-bound result tables, panel membership, the Phase 7B manifest and reproducibility instructions are released in the CrossPiezo repository.

6 Conclusion

CrossPiezo shows that MP and JARVIS agree on broad promising regions of piezoelectric chemical space while failing to reproducibly resolve the elite tail. We recommend that future high-throughput screening reports report screening-resolution curves, chance-adjusted overlaps and robust-portfolio metrics alongside aggregate correlations.

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